🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

Divide Whole Numbers by Fractions

Lesson 6-9 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can divide a whole number by a fraction by multiplying by the reciprocal.
Language Objective: I can explain my steps using the words fraction, reciprocal, quotient, and Keep, Change, Flip.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Divide Whole Numbers by Fractions | **Whole Number** | **Fraction** | **Reciprocal** | **Keep, Change, Flip** | **Quotient**

👆 *Tap any word to see what it means and a picture.*

1. ________ — Find how many fractional groups fit in a whole number by multiplying by the reciprocal.

2. A counting number like 0, 1, 2, or 3 with no fraction or decimal part is a ________.

3. A number that shows part of a whole, like 3/4, is a ________.

4. The fraction you get by flipping a fraction upside down is its ________.

5. To divide by a fraction, I ________: keep the first number, change ÷ to ×, and flip the second fraction.

6. The answer to a division problem is the ______________ .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does dividing a whole number by a fraction help the detective?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Divide Whole Numbers by Fractions** — Find how many fractional groups fit in a whole number by multiplying by the reciprocal.
  Dividir números enteros entre fracciones — Hallar cuántos grupos fraccionarios caben en un número entero multiplicando por el recíproco.
- [ ] **Whole Number** — A counting number with no fraction or decimal, like 0, 1, 2, 3.
  Número entero — Un número de contar sin fracción ni decimal, como 0, 1, 2, 3.
- [ ] **Fraction** — A number that shows part of a whole, like 3/4.
  Fracción — Un número que muestra una parte de un todo, como 3/4.
- [ ] **Reciprocal** — A fraction turned upside down.
  Recíproco — Una fracción volteada de arriba abajo.
- [ ] **Keep, Change, Flip** — A way to divide fractions: keep the first, change ÷ to ×, flip the second.
  Mantener, cambiar, invertir — Una manera de dividir fracciones: deja la primera, cambia ÷ por ×, voltea la segunda.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The detective can set up ____ markers because 10 ÷ 1/4 = ____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Each whole pound makes 2 half-pound bags, so 4 pounds makes 8 bags.

Evidence: Students see that $4 \div 1/2 = 8$ and explain there are 2 halves in each whole pound.

Reasoning: This evidence connects the definition of divide whole numbers by fractions to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can set up ____ markers because $10 \div 1/4 =$ ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Divide Whole Numbers by Fractions
2. **Notes 2:** Whole Number
3. **Notes 3:** Fraction
4. **Notes 4:** Reciprocal
5. **Notes 5:** Keep, Change, Flip
6. **Notes 6:** Quotient
7. **Watch for this mistake:** *A common mistake in Divide Whole Numbers by Fractions is doing 'Keep, Change' but skipping the 'Flip' — students multiply by the divisor as it is instead of by its reciprocal. For example, for 6 ÷ 3/4, a student might write 6 × 3/4 = 4 1/2 and stop there, but the divisor 3/4 must be flipped to 4/3 first: 6 × 4/3 = 24/3 = 8. Before you submit, check: did I flip the second fraction (the divisor) upside down before multiplying?*
8. **Try It 1:** A. 4 1/2 pouches (4 full, half a pouch left) *— 3 ÷ 2/3 = 3 × 3/2 = 9/2 = 4 1/2. You can fill 4 full pouches with half a pouch of mix left over.*
9. **Try It 2:** A. 12 *— 9 ÷ 3/4 = 9 × 4/3 = 36/3 = 12 pans.*